

Dynamic Formulation of OT : Benamou-Brenier

$$W_2^2(\alpha, \beta) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \|v_t(x)\|^2 \rho_t(x) \, dx \, dt$$

where ρ_t, v_t : measure, vector field. Subject to:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0 \quad \text{(continuity equation)}$$

$$\rho_0 = \alpha, \rho_1 = \beta \quad \text{(boundary conditions)}$$



continuity equation

Back to Continuous Time

- Wasserstein gradient flow:

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right)$$

- This is a PDE on ρ .

- **Does this look familiar?**

- It's the **continuity equation** for the vector field $v_t(x) = - \nabla \frac{\delta \mathcal{F}}{\delta \rho}(x)$.

- Takeaway:

- JKO scheme $\rightarrow$ continuous gradient flow.
- The “direction of steepest descent” in Wasserstein space is encoded by first variation of $\mathcal{F}$

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Examples of Energies & PDEs

Key power of Wasserstein grad flows: **they recover/unify many well-known PDEs**